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CivicSense: AI Based Citizen Analyzer of Citizen Complaint

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1. Abstract:

CivicSense is an automated computer software package designed to optimize municipal grievance redressal by converting unstructured citizen complaints into actionable governance intelligence. Traditional complaint systems rely on manual triage, leading to misclassification, duplicate entries and delayed response times. CivicSense addresses this through a proprietary, end-to—end six-stage processing pipeline: centralized data ingestion (via web forms and REST APIs), linguistic preprocessing with automated sequence-matching duplicate detection, machine learning classification, explainable multi-signal priority scoring, weekly digest summarization and automated administrative report generation. The classification engine utilizes TF-IDF feature extraction paired with a Linear Support Vector Classifier (LinearSVC) to route complaints into five specific municipal categories: Roads, Electricity, Water Supply, Waste Management and Public Safety. Incoming grievances are simultaneously evaluated by a deterministic priority engine that calculates an explainable urgency score based on hazard keyword severity, base category risk, emotive sentiment and temporal frequency within the reporting ward. High volumes of complaint records are compressed into structured weekly digests and exported on demand as formatted PDF or Excel reports via a local, offline-capable Flask web dashboard. Authored primarily in Python 3.12 utilizing Flask, scikit-learn, NLTK and SQLite, this registration covers the underlying source code, algorithmic scoring logic, pipeline architecture and associated web interfaces of the CivicSense platform.

Introduction: Municipalities face delayed, reactive governance because existing platforms merely collect, rather than categorize, the hundreds of thousands of unstructured daily complaints they receive. To solve this, CivicSense combines NLP and Generative AI to automatically prioritize and summarize these civic issues, transforming noisy data into actionable complaint intelligence for proactive, data-driven smart-city management.

Objectives:

1. Build an end-to-end complaint-analysis platform implementing all six stages of the paper's pipeline.
2. Achieve high, honestly-measured classification accuracy across the five service categories (Roads, Electricity, Water Supply, Waste Management, Public Safety).
3. Assign explainable High / Medium / Low priorities using the same signals the paper describes (urgency wording, category risk, emotive language, same-ward frequency, age).
4. Generate weekly narrative digests in the paper's style and export them as PDF and Excel reports.
5. Provide a dashboard that gives officers at-a-glance insight (category mix, priority mix, daily inflow, ward hotspots, trending keywords).
6. Validate everything with an automated end-to-end test suite.

Scope:

The project covers the complete software implementation: data model, NLP pipeline, trained classifier, priority engine, digest generator, report export, dashboard and tests. It uses SQLite as the central database (the paper suggests PostgreSQL/MongoDB for production scale) and a curated synthetic demonstration dataset, processed through the *real* pipeline so every statistic reflects genuine model behaviour. Live integration with external complaint portals, multilingual support and geospatial mapping are out of scope for this version and are listed as future work.

Literature Review

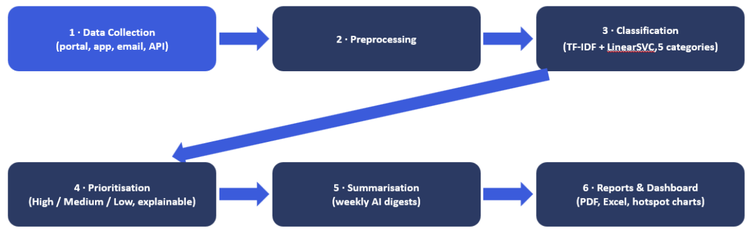
Sr. No	Paper Title
1	Centralised Public Grievance Redress and Monitoring System (PIB, 2026)
2	Classification of User Complaints and Requests in Customer (EPESS, 2025)
3	AI-Driven Citizen Grievance Redressal System (IJERT, 2026)
4	Smart Cities: A Systematic Review of Emerging Technologies (MDPI, 2026)
5	Geospatial Sentiment Analysis Using Twitter Data on Natural Disasters (ResearchGate, 2026)

Conclusion:

This project delivered a complete, working implementation of the CivicSense proposal. A six-stage AI pipeline was designed and built - centralised data collection, linguistic preprocessing with duplicate detection, TF-IDF+SVM classification across the paper's five categories, an explainable High/Medium/Low-priority engine driven by the paper's four signals, generative-style weekly digests and PDF/Excel reporting -all exposed through an officer dashboard and a REST API. The system was validated end-to-end: all 25 automated checks pass; the classifier reaches **82.2% ± 5.5%** cross-validated accuracy with every base-paper example classified correctly; priorities distribute realistically (68/94/22); duplicates are caught at a believable 16% rate; and weekly digests reproduce the paper's exact narrative style, including ward-level highlights and resource-allocation suggestions. The development process also produced genuine engineering lessons -corpus quality and metric honesty matter as much as model choice, as demonstrated by the three significant defects found and fixed during testing. CivicSense shows how NLP and Generative AI can turn the citizen's voice into actionable governance intelligence, moving municipalities from reactive complaint handling to proactive, data-driven urban management -exactly the vision the base paper sets out.

Methodology: The AI CivicSense platform streamlines public grievance management through an intelligent, end-to-end workflow. Citizens submit multimodal complaints (text, audio, image, video) that the system centrally stores, preprocesses and analyzes using NLP and machine learning. The platform automatically classifies these issues, flags duplicates and suspicious entries, assigns priority levels and routes them to the appropriate government departments. While citizens track their complaint status in real time, authorities utilize generative AI summaries, dynamic dashboards and location-based hotspot analysis to efficiently resolve issues, allocate resources and make data-driven decisions.

Workflow Diagram



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